

Cosmological Growth Constraints on Λ -Locked Discrete Entropic Gravity

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Abstract

We test the cosmological viability of a discrete entropic gravity operator exhibiting Newtonian and MOND-like regimes with Λ -locked infrared screening [1].

The effective gravitational coupling $G_{\text{eff}}(k, a)$ derived from the discrete operator is embedded into the linear growth equation for matter perturbations. We compare three scenarios: (i) Λ -locked transition at super-Hubble scale, (ii) transition at 10 Mpc, and (iii) transition at 1 Mpc.

We show that only Λ -locking preserves realistic structure growth ($\sigma_8 \approx 0.8$). Any sub-Hubble transition scale leads to catastrophic overgrowth ($\sigma_8 \gg 1$), establishing Λ -locking as a necessary condition for cosmological consistency.

1 Introduction

Modified gravity models must satisfy two independent constraints: local weak-field phenomenology (e.g., MOND-like behaviour) and cosmological structure formation consistency. These are generically in tension: nonlinear infrared enhancement of gravity increases linear growth, risking overproduction of structure.

We test whether the discrete entropic gravity operator described in [1] is compatible with observed growth. That operator produces a convergent prefactor $C \approx 2.32$ and a Λ -proportional transition scale. Here we examine whether the transition scale is cosmologically safe.

2 Effective Gravitational Coupling

From the graph operator, the deep-MOND regime produces an enhancement

$$g = C\sqrt{a_0 g_N}, \quad (1)$$

with measured $C \approx 2.32$ [1]. In Fourier space, we parameterise the scale-dependent effective coupling as

$$\frac{G_{\text{eff}}(k)}{G} = 1 + \frac{C^2 - 1}{1 + (k/k_\Lambda)^2}, \quad (2)$$

where

$$k_\Lambda = \frac{1}{L_\Lambda}, \quad L_\Lambda \propto \Lambda^{-1/2}. \quad (3)$$

This is a Lorentzian window capturing the Λ -locked screening: for $k \gg k_\Lambda$ (sub- L_Λ scales), $G_{\text{eff}} \rightarrow G$ and gravity is Newtonian. For $k \ll k_\Lambda$, the full infrared enhancement operates.

3 Linear Growth Equation

Matter perturbations δ obey

$$\delta'' + \left(2 + \frac{H'}{H}\right) \delta' - \frac{3}{2} \Omega_m(a) \frac{G_{\text{eff}}(k)}{G} \delta = 0, \quad (4)$$

where primes denote derivatives with respect to $\ln a$. We adopt Planck 2018 background parameters ($\Omega_m = 0.315$, $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\sigma_{8,\Lambda\text{CDM}} = 0.811$) from [2].

4 Scenarios Tested

We evaluate three choices of transition scale to stress-test cosmological viability.

4.1 Scenario A: Λ -Locked Transition

The Λ -locked scale is

$$L_\Lambda \sim 4.4 \text{ Gpc}, \quad k_\Lambda \sim 0.0014 \text{ h/Mpc}. \quad (5)$$

All structure formation scales ($k \gtrsim 0.01 \text{ h/Mpc}$) remain screened: $G_{\text{eff}} \approx G$.

4.2 Scenario B: 10 Mpc Transition

$$L = 10 \text{ Mpc}, \quad k_L = 0.1 \text{ h/Mpc}. \quad (6)$$

The enhancement is active across the full range of linear growth scales.

4.3 Scenario C: 1 Mpc Transition

$$L = 1 \text{ Mpc}, \quad k_L = 1.0 \text{ h/Mpc}. \quad (7)$$

Extreme stress test: enhancement penetrates into the quasi-nonlinear regime.

5 Results

Model	$\sigma_8(z=0)$	γ
ΛCDM	0.811	0.554
Graph-AQUAL (Λ -locked)	0.814	0.554
Graph-AQUAL (10 Mpc)	367	0.541
Graph-AQUAL (1 Mpc)	24 098	0.536

The Λ -locked scenario produces $\sigma_8 = 0.814$, indistinguishable from ΛCDM at the 0.4% level. Both sub-Hubble scenarios produce explosive growth by many orders of magnitude.

6 Interpretation

The results establish a sharp dichotomy:

- Λ -locking preserves linear growth. The 0.4% excess in σ_8 is well within current observational uncertainties ($\sigma_8 = 0.811 \pm 0.006$).
- Sub-Hubble transition scales produce catastrophic overgrowth. At $L = 10 \text{ Mpc}$, σ_8 exceeds unity by a factor of ~ 450 ; at $L = 1 \text{ Mpc}$, by a factor of $\sim 30\,000$.

- The discrete infrared enhancement must be globally screened at or above the Hubble scale.

Cosmological viability therefore requires

$$L_\Lambda \gtrsim H_0^{-1}. \quad (8)$$

This is not a tuned condition: it follows directly from setting the transition scale proportional to $\Lambda^{-1/2}$, which yields $L_\Lambda \sim 4.4 \text{ Gpc} \approx H_0^{-1}$. The near-coincidence is a consequence of Λ -locking, not a free parameter.

7 Discussion

The result demonstrates a strong constraint on any modified gravity model with infrared enhancement: the enhancement cannot operate on structure-formation scales without violating σ_8 bounds. This is not specific to the graph-AQUAL operator; it applies generically to models where $G_{\text{eff}} > G$ below some transition scale [3].

The discrete entropic model survives because the transition scale is set by Λ , not by a phenomenological choice. This makes Λ -locking a structural prediction of the framework, not a tuning strategy.

We note that the growth index γ is essentially unchanged across all scenarios. This is because γ characterises the shape of $f(\Omega_m)$, which depends on the background expansion, not on the amplitude of G_{eff} .

8 Limitations

This analysis has several known limitations:

- Linear growth only; no nonlinear corrections or N -body verification.
- No CMB angular power spectrum computed.
- No full Boltzmann integration (e.g., via CLASS or CAMB).
- The effective parameterisation of $G_{\text{eff}}(k)$ is a Lorentzian approximation to the full discrete operator.
- The Poisson-channel modification ($\mu(a) < 1$) is not activated; all effects enter through the background coupling.

Each of these represents a distinct avenue for future work.

9 Conclusion

We have shown that the discrete entropic gravity operator remains cosmologically viable only under Λ -locked screening at super-Hubble scale. Any sub-Hubble activation of the infrared sector leads to explosive growth, ruling out transition scales below $\sim H_0^{-1}$.

This establishes Λ -locking as a necessary condition for cosmological consistency of any infrared-enhanced gravity model. For the discrete entropic framework, Λ -locking is not imposed but derived from the bulk entropic reservoir term proportional to Λ [1].

References

- [1] M. Magnusson, *Discrete Entropic Gravity on a Cubic Graph: Emergent Newtonian and MOND Regimes with Λ -Locked Screening* (2026). doi:10.6084/m9.figshare.31348411
- [2] Planck Collaboration, *Planck 2018 results. VI. Cosmological parameters*, *Astronomy & Astrophysics* **641**, A6 (2020). doi:10.1051/0004-6361/201833910
- [3] T. Clifton, P. G. Ferreira, A. Padilla, and C. Skordis, *Modified Gravity and Cosmology*, *Physics Reports* **513**, 1–189 (2012). doi:10.1016/j.physrep.2012.01.001